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MODULE – 1

Contents:

- Introduction to Cybercrime:
- Cybercrime: Definition and Origins of the Word,Cybercrime and Information Security,
- Who are Cybercriminals? Classifications of Cybercrimes,
- An Indian Perspective, Hacking and Indian Laws., Global Perspectives

Text Book:

Sunit Belapure and Nina Godbole, "Cyber Security: Understanding Cyber Crimes, Computer Forensics And Legal Perspectives", Wiley India Pvt Ltd, ISBN: 978-81- 265-21791, 2011, First Edition (Reprinted 2018)

WeblinksandVideoLectures(e-Resources):

- https://www.youtube.com/watch?v=yC_hFm0BX28&list=PLxApjaSnQGi6Jm7LLSxvmNQjS_rt9swsu
- https://www.youtube.com/watch?v=nzZkKoREEGo&list=PL9ooVrP1hQOGPQVeapGsJCktzIO4DtI4_
- https://www.youtube.com/watch?v=6wi5DI6du4&list=PL_uaeekrhGzJlB8XQBxU3z__hDwT95xlk
- https://www.youtube.com/watch?v=KqSqyKwVuA8

1.1.1 Introduction:

The internet in India is growing rapidly. It has given rise to new opportunities in every field we can think of be it entertainment, business, sports or education. There're two sides to acoin. Internet also has its own disadvantages is Cybercrime- illegal activity committed on the internet.

Cybercrime is not a new phenomenon, the first recorded cybercrime took place in the year 1820.In 1820, Joseph Marie Jacquard, a textile manufacturer in France, produced the loom. This device allowed the repetition of a series of steps in the weaving of special fabrics. This resulted in a fear amongst Jacquard's employees that their traditional employment and livelihood were being threatened. They committed acts of sabotage to discourage Jacquard from further use of the new technology. **This is the first recorded cyber crime!**

1.2 Cybercrime: Definition and Origins of the Word

Definitions:

1.2.1 : Cybercrime

cybercrime, also called computer crime, the use of a computer as an instrument to further illegal ends, such as committing fraud, trafficking in child pornography and intellectual property, stealing identities, or violating privacy. Cybercrime, especially through the Internet,has grown in importance as the computer has become central to commerce, entertainment, and government.

OR

Cybercrime (computer crime) is any illegal behavior, directed by means of electronic operations, that targets the security of computer systems and the data processed by them.

1.2.2 : Cyberspace

Cyberspace refers to the virtual computer world and more specifically, an electronic medium that is used to facilitate online communication. Cyberspace typically involves a large computer network made up of many worldwide computer subnetworks that employ TCP/IP protocol to aid in communication and data exchange activities.

Cyberspace is most definitely a place where you chat, explore, research and play

1.2.3 : Cybersquatting

The term cybersquatting refers to the unauthorized registration and use of Internet domain names that are identical or similar to trademarks, service marks, company names or personal names. Cybersquatting registrants obtain and use the domain name with the bad faith intent to profit from the goodwill of the actual trademark owner. Both the federal government and the Internet Corporation for Assigned Names and Numbers have taken action to protect the owners of trademarks and businesses against cybersquatting abuses

1.2.4 : Cyberpunk

A programmer who breaks into computer systems in order to steal or change or destroy information as a form of cyber-terrorism. cyber-terrorist, hacker.

1.2.5 : Cyberwarfare

Cyber warfare is usually defined as a cyber-attack or series of attacks that target a country. It has the potential to wreak havoc on government and civilian infrastructure and disrupt critical systems, resulting in damage to the state and even loss of life

OR

Cyber warfare involves the actions by a nation-state or international organization to attack and attempt to damage another nation's computers or information networks through, for example, computer viruses or denial-of-service attacks.

1.2.6 : Cyberterrorism

Cyberterrorism is the premeditated, politically motivated attack against information, computer systems, programs and data which result in violence against noncombatant targets by sub national groups or clandestine agents. Attackers often do this by damaging or disrupting critical infrastructure.

OR

Any person, group or organization who, with terrorist intent, utilizes accesses or aids in accessing a computer or computer network or electronic system or electronic device by any available means and there by knowingly engages in or attempts to engage in a terrorist act commits the offence of cyberterrorism.

1.2.7 : Cybercrime

Cybercrime (computer crime) is any illegal behavior, directed by means of electronic operations, that targets the security of computer systems and the data processed by them.

There are two types of attack are prevelent in cybercrime

- Techno crime: A premeditated act against a system or systems, with the intent to copy, steal, prevent access, corrupt or otherwise deface or damage parts of or the complete computer system.
- Techno-vandalism: Techno-vandalism occurs when unauthorized access to a computer results in damage to files or programs, not so much for profit but for the challenge. In such cases, the damage or loss may be intentional or accidental

Internet is one of the means by which the offenders can gain priced sensitive information of companies, firms, individuals, banks and can lead to intellectual property (IP) crimes (such as stealing new product plans, its description, market program plans, list of customers, etc.), selling illegal articles, pornography/child pornography, etc. This is done using methods such as Phishing, Spoofing, Pharming, Internet Phishing, wire transfer, etc. and use it to their own advantage without the consent of the individual. "Phishing" refers to an attack using mail programs to deceive or coax Internet users into disclosing confidential information that can be then exploited for illegal purposes. Figure 1.2 shows the increase in Phishing hosts.

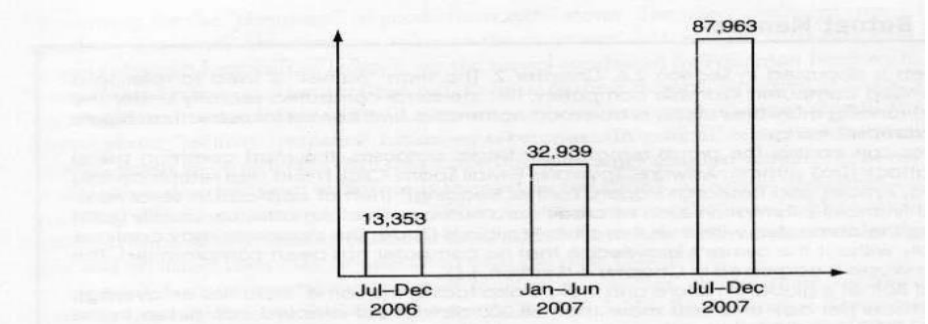

Figure 1.2 | Rise in the number of Phishing hosts.

1.3 : Cybercrime and information security

- Lack of information security give rise to cybercrime

Cyber Security:

Cyber security means protecting information, equipment, devices, computer, computer resource, communication device and information stored therein from unauthorized access, use, disclosure, disruption, modification or destruction

Challenges for Securing Data in Business Perspective:

- Cybercrime occupy an important space in information security due to their impact.
- Most organizations do not incorporate the cost of the vast majority of computer security incidents into their accounting
- The difficulty in attaching a quantifiable monetary value to the corporate data and yet corporate data get stolen/lost
- Financial loses may not be detected by the victimized organization in case of Insider attacks: such as leaking customer data

The Botnet Menace

The term Botnet is used to refer to a group of compromised computers (zombie computers ie.. Personal computers secretly under the control of hackers) running malwares under a common command and control infrastructure.

Figure 1.3 shows how a "Zombie" works

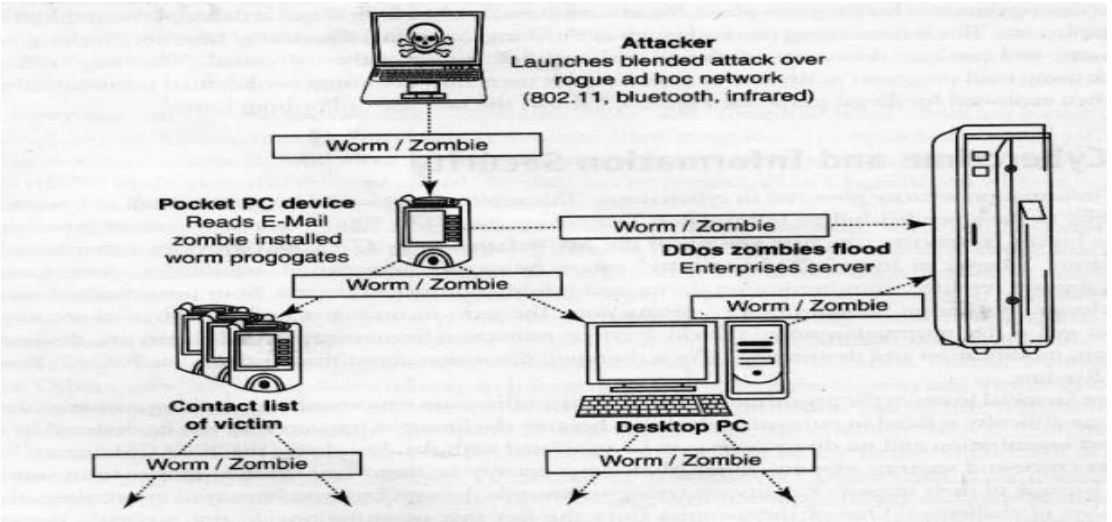

Figure 1.3 | How a zombie works.

Cybercrime trends over years

Table 1	2004	2005	2006	2007	2008
Denial of service	39%	32%	25%	25%	21%
Laptop theft	49%	48%	47%	50%	42%
Telecom fraud	10%	10%	8%	5%	5%
Unauthorized access	37%	32%	32%	25%	29%
Virus	78%	74%	65%	52%	50%
Financial fraud	8%	7%	9%	12%	12%
Insider abuse	59%	48%	42%	59%	44%
System penetration	17%	14%	15%	13%	13%
Sabotage	5%	2%	3%	4%	2%
Theft/loss of proprietary info	10%	9%	9%	8%	9%
from mobile devices					4%
from all other sources					5%
Abuse of wireless network	15%	16%	14%	17%	14%
Web site defacement	7%	5%	6%	10%	6%
Misuse of Web application	10%	5%	6%	9%	11%
Bots				21%	20%
DNS attacks				6%	8%
Instant messaging abuse				25%	21%
Password sniffing				10%	9%
Theft/loss of customer data				17%	17%
from mobile devices					8%
from all other sources					8%

Several categories of incidences:

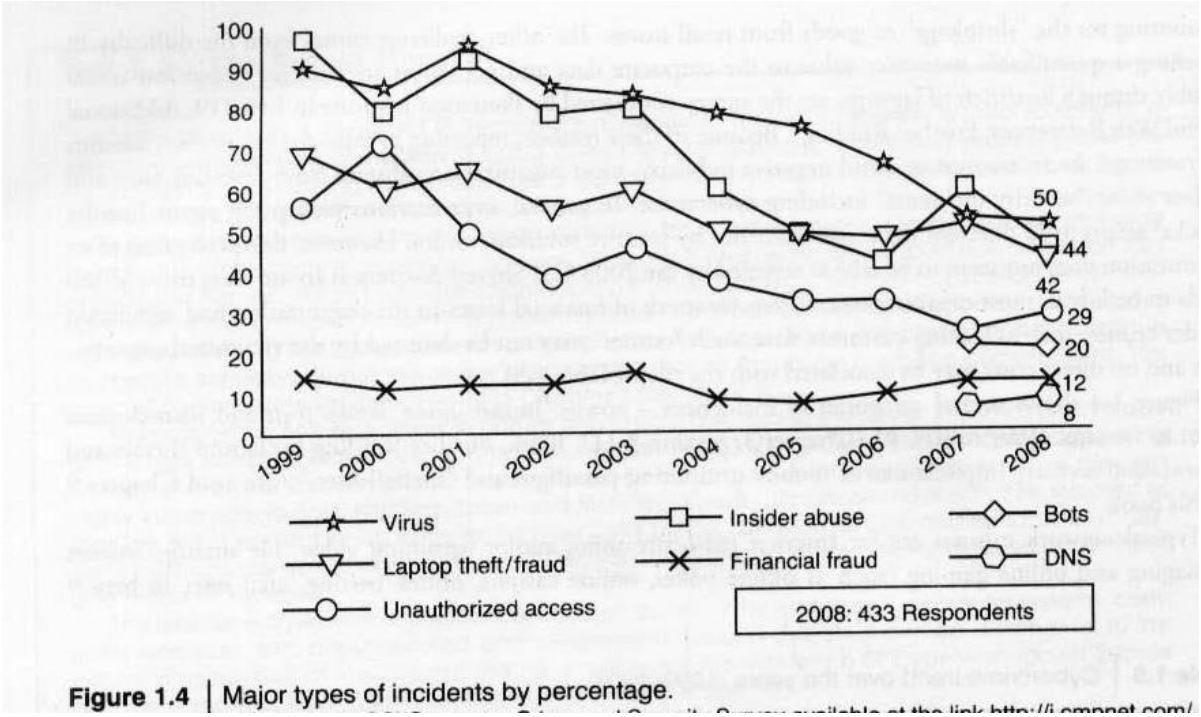

Figure 1.4 | Major types of incidents by percentage.

1.4 : Who are Cybercriminals?

Cybercrime involves such activities as child pornography, credit card fraud, cyberstalking, defaming another online, gaining unauthorized access to computer systems, ignoring copyright, software licensing and trademark protection, overriding encryption to make illegal copies, software piracy and stealing another's identity to perform criminal acts.

They can be categorized into three groups.

- Type 1: Cybercriminals- hungry for recognition
  - Hobby hackers
    - A person who enjoys exploring the limits of what is possible, in a spirit of playful cleverness. May modify hardware/ software
  - IT professional (social engineering):
    - Ethical hacker
  - Politically motivated hackers :
    - promotes the objectives of individuals, groups or nations supporting a variety of causes such as: Anti-globalization, transnational conflicts and protest
  - Terrorist organizations
    - Cyberterrorism

- Use the internet attacks in terrorist activity
- Large scale disruption of computer networks , personal computers attached to internet via viruses

Type 2: Cybercriminals- not interested in recognition

- Psychological perverts
- Financially motivated hackers (Corporate espionage)
- State – Sponsored hacking (National espionage, Sabotage)
- Organized Criminals.

Type 3: Cybercriminals- the insiders

- Disgruntled or former employees seeking revenge
- Competing companies using employees to gain economic advantage through damage and/ or theft.

1.5 : Classification of Cyber Crimes

Cybercrimes can be classified in to 5 major categories as the following:

(1) Cybercrime against Individual

(2) Cybercrime Against Property

(3) Cyber crime Against Organization

(4) Cyber crime Against Society

(5) Crimes emanating from Usenet newsgroup

1.5.1 Cybercrime Against Individuals

(i) Email spoofing and other online frauds:

A spoofed email is one in which the e-mail header is forged so that the mail appears to originate from one source but actually has been sent from another source.

Ex: roopa@asianlaws.org

Spoofs her E-mail and sends obscene/vulgar messagesto all her acquaintrances.

(ii) Phishing :

Phishing attacks are the practice of sending fraudulent communications that appear to come from a reputable source. It is usually done through email. The goal is to steal sensitive data

like credit card and login information, or to install malware on the victim machine. Phishing is a common type of cyber attack that everyone should learn about in order to protect themselves

(iii) Spamming:

Spamming is the use of electronic messaging systems like e-mails and other digital delivery systems and broadcast media to send unwanted bulk messages indiscriminately. The term spamming is also applied to other media like in internet forums, instant messaging and mobile text messaging, social networking spam, junk fax transmissions, television advertising and file sharing network spam, web search engine spam, video sharing sites etc.

Those who continually attempt to subvert or Spam the search engines may be permanently excluded from the search index. Therefore the following web publishing techniques should be avoided.

- Repeating keywords
- Use of keywords that do not relate to the content of the site
- Use of fast meta refresh
- Redirection
- IP Cloaking
- Use of colored text on the same color backgrounds
- Tiny text usage
- Duplication of pages with different URLs
- Hidden links
- Use of different pages that bridge to the same URL

(iv) Cyber Defamation :

This occurs when defamation takes place with the help of computers and/or the Internet. E.g. someone publishes defamatory matter about someone on a website or sends e-mails containing defamatory information.

According to the IPC section 499:

- Explanation 1.—It may amount to defamation to impute anything to a deceasedperson, if the imputation would harm the reputation of that person if living, and is intended to be hurtful to the feelings of his family or other near relatives.
- Explanation 2.—It may amount to defamation to make an imputation concerning a company or an association or collection of persons as such.
- Explanation 3.—An imputation in the form of an alternative or expressed ironically, may amount to defamation.
- Explanation 4.—No imputation is said to harm a person's reputation, unless that imputation directly or indirectly, in the estimation of others, lowers the moral or intellectual character of that person, or lowers the character of that person in respect of his caste or of his calling, or lowers the credit of that person, or causes it to be believed that the body of that person is in a loathsome state, or in a state generally considered as disgraceful.

(v) Harassment & Cyber Stalking:

Cyber Stalking Means following an individual's activity over internet. It can be done with the help of many protocols available such as e- mail, chat rooms, user net groups.

(vi) Computer sabotage:

Computer sabotage is the input, alteration, erasure or suppression of computer data or computer programmes, or interference with computer systems, with the intent to hinder the functioning of a computer or a telecommunication system through the introduction of worms,viruses or logic bombs.

- It can be used to gain economic advantage over a competitor to promote the illegal activities of terrorists or to steal data or programs for extortion purposes.
- Logic bombs are event – dependent programs created to do something only when a certain event occurs.
- Some viruses may be termed as logic bombs because they lie dormant all through the year and become active only on a particular date.

(vii) Pornographic offenses:

The term "pornography" can be defined as the reporting or portrayal of sexual actions in order to produce sexual excitement through books, films, or other media. Pornographic websites, pornographic material created using computers, and the use of the internet to download and transmit pornographic films, texts, photographs, and photos, among other things, fall under this category.

- Pedophiles are people who physically or psychologically coerce minors to engage in sexual activities, which the minors would not consciously consent to. Here is how pedophilies operate:

Step1: Pedophiles use a false identity to trap the children/teenages

Step2: They seek children/teens in the kids areas on the services, such as the Teans BB,Games BB or chat areas where the children gather.

Step3: They befriend children/teens

Step4: They extract personal information from the child/teen by winning his/her confidence.

Step5: Pedophiles get E-mail address of the child/teen and start making contacts on the victim's E-Mail address as well. Sometimes, these E-Mails contain sexually explicit language.

Step6: They start sending pornographic images/text to the victim including child pornographic images in order to help child/teen shed his/her inhibitions so that a feeling is created in the mind of the victim that what is being fed to him is normal and that everybody does it.

Step7: The pedophiles set up a meeting with the child/teen out of the house and then drag him/her into the net to further sexually assault him/her or to use him/her as a sex object.

(Viii) Password sniffing:

Password Sniffing is a hacking technique that uses a special software application that allows a hacker to steal usernames and passwords simply by observing and passively

recording network traffic. This often happens on public WiFi networks where it is relatively easy to spy on weak or unencrypted traffic.

Table 1.6 | Classifying cybercrimes – broad and narrow

	Cybercrime in Narrow Sense	*Cybercrime in Broad Sense*	
Role of computer	*Computer as an object* The computer/information stored on the computer is the subject/target of the crime	*Computer as a tool* The computer/or information stored on the computer constitutes an important tool for committing the crime	*Computer as the environment or context* The computer/information stored on the computer plays a non-substantial role in the act of crime, but does contain evidence of the crime
Examples	Hacking, computer sabotage, DDoS-attacks (distributed denial-of-service attacks), virtual child pornography	Computer fraud, forgery distribution of child pornography	Murder using computer techniques, bank robbery and drugs trade

1.5.2 Cybercrime Against Property

i) Credit card frauds:

Credit card fraud refers to using a credit card to obtain money or goods fraudulently. Thieves may steal a credit card, copy the number off a credit card, or take over a victim's account and have the credit card mailed to their (the criminal's) address. They may also open a new credit card in the victim's name or try a variety of other techniques to steal money or buy assets.

ii) Intellectual Property (IP) Crimes:

Intellectual property crime is committed when someone manufactures, sells or distributes counterfeit or pirated goods, such as patents, trademarks, industrial designs or literary and artistic works, software piracy, theft of computer source code for commercial gain.

iii) Internet time theft:

The person who gets access to someone else's ISP(Internet Service Provider) user ID and password, either by hacking or by gaining access to it by illegal means, uses it to access the Internet without the other person's knowledge.

1.5.3 Cybercrime Against Organization

- Unauthorized accessing of computer
- Password sniffing
- Denial-of-service attacks
- Virus attack/dissemination of viruses
- E-mail bombing/mail bombs
- Salami attack/Salami technique
- Logic bomb
- Trojan Horse
- Data diddling
- Crimes emanating from Usenet newsgroup
- Industrial spying/industrial espionage
- Computer network intrusions
- Software piracy

Password sniffing:

Password sniffing is an attack on the Internet that is used to steal user names and passwords from the network.

Denial-of-service attacks:

A Denial-of-Service (DoS) attack is an attack meant to shut down a machine or network, making it inaccessible to its intended users. DoS attacks accomplish this by flooding the target with traffic, or sending it information that triggers a crash.

Victims of DoS attacks often target web servers of high-profile organizations such as banking, commerce, and media companies, or government and trade organizations. Though DoS attacks do not typically result in the theft or loss of significant information or other assets, they can cost the victim a great deal of time and money to handle

Virus attack/dissemination of viruses:

A computer virus is malicious code that replicates by copying itself to another program, computer boot sector or document and changes how a computer works. A virus spreads

between systems after some type of human intervention. Viruses replicate by creating their own files on an infected system, attaching themselves to a legitimate program, infecting a computer's boot process or infecting user documents. The virus requires someone to knowingly or unknowingly spread the infection. In contrast, a computer worm is standalone programming that does not require human interaction to spread. Viruses and worms are two examples of malware, a broad category that includes any type of malicious code.

E-mail bombing/mail bombs:

A very large number of emails to the victim to crash victim's E-mail account or to make victim's mail servers crash. Computer programs can be written to instruct a computer to do such tasks on a repeated basis. Terrorism has hit the internet in the form of mail bombings. By instructing a computer to repeatedly send E-Mail to a specified person's E-Mail address, the cybercriminal can overwhelm the recipient's personal account and potentially shut down entire systems.

Salami attack/Salami technique:

A salami attack is a type of cybercrime that attackers typically use to commit financial crimes. Criminals steal money or resources from financial accounts on a system one at a time.This attack occurs when several minor attacks combine to form a powerful attack. Because ofthis type of cybercrime, these attacks frequently go undetected.

Ex: A bank employee inserts a program, into the bank's servers, that deducts a small amount of money (say ₹2/- or a few cents in a month) from the account of every customer. No account holder will probably notice this unauthorized debit, but the bank employee will makea sizable amount every month.

Logic bomb:

- It can be used to gain economic advantage over a competitor to promote the illegal activities of terrorists or to steal data or programs for extortion purposes.
- Logic bombs are event – dependent programs created to do something only when a certain event occurs.
- Some viruses may be termed as logic bombs because they lie dormant all through the year and become active only on a particular date.

Trojan Horse:

Trojan Horse is a program in which malicious or harmful code is contained inside apparently harmless programming or data in such a way that it can get control and cause harm.

Ex: Ruining the file allocation table on the hard disk.A Trojan Horse may get widely redistributed as part of a computer virus. The term Trojan Horse comes from Greek mythology about the Trojan War.

Data diddling:

- Data diddling involves changing data input in a computer.
- In other words, information is changed from the way it should be entered by a person typing in the data.
- Usually, a virus that changes data or a programmer of the database or application has pre-programmed it to be changed**.**

- For example, a person entering accounting may change data to show their account, or that or a friend or family member, is paid in full. By changing or failing to enter the information, they are able to steal from the company.
- To deal with this type of crime, a company must implement policies and internal controls.
- This may include performing regular audits, using software with built-in features to combat such problems, and supervising employees.

Crimes emanating from Usenet newsgroup/Newsgroup Spam:

Usenet convention defines spamming as "excessive multiple posting (EMP)", that is, the repeated posting of a message (or substantially similar messages). The first widely recognized Usenet spam (though not the most famous) was posted on 18 January 1994 by

Clarence L. Thomas IV, a sysadmin at Andrews University.[Entitled "Global Alert for All: Jesus is Coming Soon", it was a fundamentalist religious tract claiming that "this world's history is coming to a climax." The newsgroup posting Bot Serdar Argic also appeared in early 1994, posting tens of thousands of messages to various newsgroups, consisting of identical copies of a political screed relating to the Armenian genocide.

Industrial spying/industrial espionage:

- Industrial espionage is the covert and sometimes illegal practice of investigating competitors to gain a business advantage.
- The target of investigation might be a trade secret such as a proprietary product specification or formula, or information about business plans.
- In many cases, industrial spies are simply seeking any data that their organization can exploit to its advantage.

Computer network intrusions:

- An intrusion to computer network from anywhere in the world and steal data, plant viruses, create backdoors, insert trojan horse or change passwords and user names.
- An intrusion detection system (IDS) inspects all inbound and outbound networkactivity and identifies suspicious patterns that may indicate a network or system attack from someone attempting to break into or compromise a system.
- The practice of strong password is therefore important

Software piracy:

- Theft of software through the illegal copying of genuine programs or the counterfeiting and distribution of products intended to pass for the original.

Examples:

- End-user copying- Friends loaning disks to each other or organizations under reporting the number of software installations they have made or organizations not tracking their software licences
- Hard disk loading with illicit means – Hard disk vendors load pirated software
- Counterfeiting – Large scale duplication and distribution of illegally copied software

- Illegal downloads from internet – By intrusion, by cracking serial numbers

1.5.4 Cybercrime Against Society

- Forgery
- Cyberterrorism
- Web Jacking

Forgery

- The act of forging something, especially the unlawful act of counterfeiting a document or object for the purposes of fraud or deception.
- Something that has been forged, especially a document that has been copied or remade to look like the original.
- Counterfeit currency notes, postage, revenue stamps, marksheets, etc., can be forged using sophisticated computers, printers and scanners

Cyberterrorism

Cyberterrorism is the premeditated, politically motivated attack against information, computer systems, programs and data which result in violence against noncombatant targets by sub national groups or clandestine agents. Attackers often do this by damaging or disrupting critical infrastructure.

OR

Any person, group or organization who, with terrorist intent, utilizes accesses or aids in accessing a computer or computer network or electronic system or electronic device by any available means and thereby knowingly engages in or attempts to engage in a terrorist act commits the offence of cyberterrorism.

Web Jacking:

- This term is derived from the term hi jacking.
- In these kinds of offences, the hacker gains access and control over the web site of another.
- He may even change the information on the site.

- The first stage of this crime involves “password sniffing”.
- The actual owner of the website does not have any more control over what appears on that website
- This may be done for fulfilling political objectives or for money

Web Jacking - Real life examples

- Recently the site of MIT (Ministry of Information Technology) was hacked by the Pakistani hackers and some obscene matter was placed therein.
- Further the site of Bombay crime branch was also web jacked.
- Another case of web jacking is that of the ‘gold fish’ case. In this case the site was hacked and the information pertaining to gold fish was changed.

1.5.5 Crimes emanating from Usenet newsgroup

- Usenet is a popular means of sharing and distributing information on the Web with respect to specific topics or subjects.
- Usenet is a mechanism that allows sharing information in a many-to-many manner.
- There is no technical method available for controlling the contents of any newsgroup.
- Possible to put Usenet to following criminal use:
  1. Distribution/sale of pornographic material
  2. Distribution/sale of pirated software packages.
  3. Distribution of hacking software.
  4. Sale of stolen credit card numbers.
  5. Sale of stolen data/stolen property

Hacking:

The Purposes of hacking are many, the main ones are as follows.

1.Greed

2. Power

3. Publicity

4. Revenge

5.Adventure

6.Desire to access forbidden information

7.Destructive mindset

Every act committed toward breaking into a computer and/ or network is hacking. Hackers write or use ready-made computer programs to attack the target computer. Theypossess the desire to destruct and they get enjoyment out of such destruction. Some hackers hack for personal monetary gains, such as stealing credit card information, transferring money from various bank accounts to their own account followed by withdrawal of money. Government websites are hot on hackers target lists.

1.6 Cybercrime: The Legal Perspective

- Cybercrime possess a mammoth challenge
- Computer crime: Criminal Justice Resource Manual(1979)

Any illegal act for which knowledge of computer technology is essential for a successful prosecution.

- International legal aspects of computer crimes were studied in 1983

Encompasses any illegal act for which the knowledge of computer technology is essential for its perpetration

The network context of cybercrime make it one of the most globalized offenses of the present and most modernized threats of the future.

Solution:

1. Divide information system into segments bordered by state boundaries.

- Not possible and unrealistic because of globalization

2. Or incorporate the legal system into an integrated entity obliterating these state boundaries.

1.7 Cybercrimes: An Indian Perspective

- India has the fourth highest number of internet users in the world.
- 45 million internet users in India
- 37% of all Internet access happen from cybercafes

- 57% Indian Internet users are between 18 and 35 years
- The Information Technology (IT) Act, 2000, specifies the acts which are punishable. Since the primary objective of this Act is to create an enabling environment for commercial use of I.T.
- The majority of offenders were under 30 Years.
- The maximum cybercrime cases, about 46%, were related to incidents of cyberporonography, followed by hacking.
- In over 60% of these cases, offenders were between 18 and 30 years, according to the crime in 2007
- 217 cases were registered under IT Act during the year 2007 as compared to 142 cases during the previous year (2006)
- Thereby reporting an increase of 52.8% in 2007 over 2006.
- 22.3% cases (49out of 217 cases) were reported from Maharashtra followed by Karnataka (40), Kerala (38) and Andhra Pradesh and Rajasthan (16 each).

Cyber Crimes/Cases Registered and Persons Arrested under IT Act during 2004-2007

SL. NO.	Crime Heads	Cases Registered				% Variation in 2007 over 2006	Persons Arrested				% Variation in 2007 over 2006
		2004	2005	2006	2007		2004	2005	2006	2007	
1	Tampering computer source documents	2	10	10	11	10.0	0	10	8	2	-75
2	Hacking with Computer System										
	i) Loss/damage to computer resource/utility	14	33	25	30	20.0	31	27	34	25	26.5
	ii)Hacking	12	41	34	46	35.3	1	14	29	23	-20.7
3	Obscene publication/transmission in electronic form	34	88	69	99	43.5	21	125	81	86	6.2
4	Failure										
	i) Of compliance/orders of Certifying Authority	0	1	0	2	-	0	0	0	1	-
	ii) To assist in decrypting the information intercepted by Govt. Agency	0	0	0	2	-	0	0	0	0	-
5	Un-authorised access/attempt to access to protected computer system	0	0	0	4	-	0	0	0	0	-
6	Obtaining licence or Digital Signature Certificate by misrepresentation/suppression of fact	0	0	0	11	-	0	0	0	11	-
7	Publishing false Digital Signature Certificate	0	0	0	0	-	0	0	0	0	-
8	Fraud Digital Signature Certificate	0	1	1	3	200.0	0	3	0	3	-
9	Breach of confidentiality/privacy	6	3	3	9	200.0	7	13	2	3	50.0
10	Other	0	0	0	0	-	0	0	0	0	-
	Total	**68**	**179**	**142**	**217**	**52.8**	**60**	**192**	**154**	**154**	**0.0**

Incidence of Cyber Crimes in Cities

- 17 out of 35 mega cities did not report any case of Cyber Crime i.e, neither under the IT Act nor under IPC Sections) during the year 2007.
- 17 mega cities have reported 118 cases under IT Act and 7 megacities reported 180 cases undervarious section of IPC.
- There was an increase of 32.6% (from 89 cases in 2006 to 118 cases in 2007) in cases under IT Act as compared to previous year (2006),
- and an increase of 26.8% (from 142 cases in 2006 to 180 cases in 2007) of cases registered under various section of IPC
- Bengaluru (40), Pune (14) and Delhi (10) cities have reported high incidence of cases (64 out of 118 cases) registered under IT Act, accounting for more than half of the cases (54.2%) reported under the Act.

1.8 A Global Perspective on Cybercrimes

- In Australia, cybercrime has a narrow statutory meaning as used in the Cyber Crime Act 2001, which details offenses against computer data and systems.

- In the Council of Europe's (CoE's) Cyber Crime Treaty, cybercrime is used as an umbrella term to refer to an array of criminal activity including offenses against computer data systems, computer-related offenses, content offenses and copyright offenses.

- August 4,2006 Announcement: The US Senate ratifies CoE Convention on Cyber Crime. The convention targets hackers, those spreading destructive computer viruses, those using the Internet for the sexual exploitation of children or the distribution of racist material and terrorists attempting to attack infrastructure facilities or financial institutions. The convention is in full accord with all the US constitutional protections, such as speech and other civil liberties, and will require no change to the US laws.

- CoE Cyber Crime Convention (1997-2001) was the first international treaty seekingto address internet crimes by harmonizing national laws, improving investigative techniques and increasing cooperation among nations.

- Private sectors expertise should be increasingly involved in the development and implementation of a country's cybersecurity strategy.